Chapter 15 — Basic Separation Techniques

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Filtration separates:

(a) Two miscible liquids **(b)** Insoluble solid from liquid **(c)** Gases only **(d)** Dissolved salt from water

2. Solid left on filter paper:

(a) Distillate **(b)** Residue **(c)** Eluent **(d)** Filtrate

3. Liquid passing through filter:

(a) Residue **(b)** Precipitate only **(c)** Filtrate **(d)** Solute only

4. Crystallization separates:

(a) Soluble solid from solution **(b)** Two gases **(c)** Metals only **(d)** Insoluble solid only

5. Crystallization solvent should dissolve solute:

(a) Hot but little cold **(b)** Cold but little hot **(c)** Never **(d)** Equally always

6. Slow cooling gives:

(a) Purer crystals **(b)** Impure mass **(c)** No crystals **(d)** Gas only

7. Simple distillation separates by:

(a) Magnetism **(b)** Density only **(c)** Boiling point **(d)** Colour

8. Distillation of salt water gives:

(a) Pure water **(b)** NaCl gas **(c)** Salt crystals only **(d)** No separation

9. Fractional distillation separates:

(a) Ions only **(b)** Gases only **(c)** Insoluble solids **(d)** Close boiling liquids

10. Fractionating column provides:

(a) Repeated vaporization/condensation **(b)** Magnetic separation **(c)** Filtration **(d)** Crystallization

11. Ethanol-water separation uses:

(a) Fractional distillation **(b)** Decantation **(c)** Paper chromatography only **(d)** Filtration

12. Petroleum refining uses:

(a) Crystallization only **(b)** Filtration **(c)** Fractional distillation **(d)** Sublimation only

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13. Chromatography separates by:

(a) Melting only **(b)** Magnetism **(c)** Differential movement **(d)** Boiling only

14. Paper chromatography stationary phase:

(a) Air **(b)** Flame **(c)** Paper/water **(d)** Metal

15. Mobile phase is:

(a) Solvent **(b)** Pencil line **(c)** Paper only **(d)** Air only

16. Rf equals:

(a) solute distance/solvent distance **(b)** solvent/solute **(c)** mass/volume **(d)** time/distance

17. Colourless spots are located by:

(a) Weighing only **(b)** Taste **(c)** Locating agents/UV **(d)** Smell only

18. Pure solid has:

(a) No melting **(b)** Sharp melting point **(c)** Broad range **(d)** Lower boiling always

19. Impurity usually:

(a) No effect **(b)** Lowers melting point **(c)** Makes gas **(d)** Raises melting point

20. Pure chromatogram shows:

(a) One spot **(b)** No solvent **(c)** Many spots **(d)** No sample

21. Distillate is:

(a) Filter paper **(b)** Condensed vapour **(c)** Original mixture **(d)** Residue

22. Condenser cools:

(a) Residue **(b)** Vapour **(c)** Sample spot **(d)** Filter paper

23. Thermometer bulb is placed at:

(a) Side-arm **(b)** In receiver **(c)** Bottom of flask **(d)** In condenser water

24. Sublimation separates:

(a) Only metals **(b)** Only liquids **(c)** Only gases **(d)** Sublimable solid

25. Insoluble impurities removed by:

(a) Hot filtration **(b)** Cooling only **(c)** Chromatography only **(d)** Distillation only

26. Solvent front must be marked:

(a) Never **(b)** Immediately **(c)** After drying days **(d)** Before run only

27. Capillary action moves solvent in:

(a) Distillation only **(b)** Titration only **(c)** Filtration only **(d)** Paper chromatography

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28. A mixture of sand and water uses:

(a) Fractional distillation **(b)** Sublimation only **(c)** Chromatography only **(d)** Filtration

29. A mixture of salt and water uses:

(a) Evaporation/distillation **(b)** Decantation only **(c)** Magnetism **(d)** Filtration only

30. Rf value is constant under:

(a) Same conditions **(b)** Any solvent **(c)** Any temperature **(d)** Any paper

31. Filtration separates:

(a) Dissolved solid only **(b)** Insoluble solid from liquid **(c)** Miscible liquids **(d)** Gases only

32. Residue is:

(a) Solid left on filter **(b)** Liquid passed through **(c)** Solvent only **(d)** Vapour

33. Filtrate is:

(a) Gas **(b)** Crystal only **(c)** Solid left on filter **(d)** Liquid passed through

34. Crystallization separates:

(a) Insoluble solid only **(b)** Miscible gases **(c)** Metals only **(d)** Soluble solid from solution

35. Good crystallization solvent dissolves solute:

(a) Poorly hot, well cold **(b)** Never **(c)** Well hot, poorly cold **(d)** Equally at all T

36. Simple distillation separates:

(a) Insoluble solids only **(b)** Liquid from dissolved solid **(c)** Gases only **(d)** Close-boiling liquids only

37. Fractional distillation separates:

(a) Insoluble solids **(b)** Metals only **(c)** Gases only **(d)** Miscible liquids with close boiling points

38. Fractionating column provides:

(a) Filtration only **(b)** Crystallization only **(c)** Repeated vaporization-condensation **(d)** No separation

39. Chromatography separates by:

(a) Colour only **(b)** Mass only **(c)** Different distribution between phases **(d)** Density only

40. Stationary phase in paper chromatography is:

(a) Solvent front **(b)** Metal plate **(c)** Air **(d)** Water held in paper

41. Mobile phase is:

(a) Solvent moving up paper **(b)** Paper only **(c)** Air only **(d)** Solute only

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42. Rf equals:

(a) solvent distance/solute distance **(b)** time/distance **(c)** solute mass/solvent mass **(d)** solute distance/solvent distance

43. Colourless spots can be located by:

(a) Weighing only **(b)** UV or locating agent **(c)** Filtering only **(d)** Boiling only

44. Pure solid has:

(a) Sharp melting point **(b)** Broad melting range **(c)** No melting point **(d)** Random melting

45. Pure liquid has:

(a) Broad boiling range always **(b)** Constant boiling point **(c)** No boiling point **(d)** Random boiling

46. Chromatography purity check gives:

(a) One spot **(b)** No stationary phase **(c)** Many spots **(d)** No solvent front

47. Hot filtration removes:

(a) Insoluble impurities **(b)** Solvent only **(c)** Dissolved solute only **(d)** Crystals only

48. Slow cooling gives:

(a) Smaller impure crystals **(b)** Gas only **(c)** Larger purer crystals **(d)** No crystals

49. Distillate is:

(a) Condensed vapour collected **(b)** Solid residue **(c)** Original mixture **(d)** Filter paper

50. Thermometer in distillation measures:

(a) Solvent colour **(b)** Vapour temperature **(c)** Solid density **(d)** Liquid mass

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Answer Key with Brief Reasons

1. **Answer: (b) Insoluble solid from liquid** — Heterogeneous mixture.
2. **Answer: (b) Residue** — Filtration term.
3. **Answer: (c) Filtrate** — Filtration term.
4. **Answer: (a) Soluble solid from solution** — Pure crystals.
5. **Answer: (a) Hot but little cold** — Crystal formation.
6. **Answer: (a) Purer crystals** — Ordered growth.
7. **Answer: (c) Boiling point** — Vaporization/condensation.
8. **Answer: (a) Pure water** — Water distillate.
9. **Answer: (d) Close boiling liquids** — Repeated condensation.
10. **Answer: (a) Repeated vaporization/condensation** — Better separation.
11. **Answer: (a) Fractional distillation** — Close boiling points.
12. **Answer: (c) Fractional distillation** — Fractions.
13. **Answer: (c) Differential movement** — Stationary/mobile phases.
14. **Answer: (c) Paper/water** — Adsorption.
15. **Answer: (a) Solvent** — Moves by capillary action.
16. **Answer: (a) solute distance/solvent distance** — Retardation factor.
17. **Answer: (c) Locating agents/UV** — Visualization.
18. **Answer: (b) Sharp melting point** — Purity check.
19. **Answer: (b) Lowers melting point** — Melting depression.
20. **Answer: (a) One spot** — Purity check.
21. **Answer: (b) Condensed vapour** — Distillation product.
22. **Answer: (b) Vapour** — Liquefies vapour.
23. **Answer: (a) Side-arm** — Vapour temperature.
24. **Answer: (d) Sublimable solid** — Solid to vapour.
25. **Answer: (a) Hot filtration** — Before crystallization.
26. **Answer: (b) Immediately** — Rf calculation.
27. **Answer: (d) Paper chromatography** — Upward movement.
28. **Answer: (d) Filtration** — Insoluble solid.
29. **Answer: (a) Evaporation/distillation** — Dissolved solid.
30. **Answer: (a) Same conditions** — Identification.
31. **Answer: (b) Insoluble solid from liquid** — Residue and filtrate.

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32. Answer: (a) Solid left on filter — Filtration term.

33. Answer: (d) Liquid passed through — Filtration term.

34. Answer: (d) Soluble solid from solution — Crystal formation.

35. Answer: (c) Well hot, poorly cold — Temperature-dependent solubility.

36. Answer: (b) Liquid from dissolved solid — Vaporization-condensation.

37. Answer: (d) Miscible liquids with close boiling points — Fractionating column.

38. Answer: (c) Repeated vaporization-condensation — Theoretical plates.

39. Answer: (c) Different distribution between phases — Stationary/mobile phases.

40. Answer: (d) Water held in paper — Paper support.

41. Answer: (a) Solvent moving up paper — Capillary action.

42. Answer: (d) solute distance/solvent distance — Retardation factor.

43. Answer: (b) UV or locating agent — Visualization.

44. Answer: (a) Sharp melting point — Purity check.

45. Answer: (b) Constant boiling point — Purity check.

46. Answer: (a) One spot — Single component.

47. Answer: (a) Insoluble impurities — Before cooling.

48. Answer: (c) Larger purer crystals — Selective lattice growth.

49. Answer: (a) Condensed vapour collected — Distillation product.

50. Answer: (b) Vapour temperature — Boiling component.

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